

Module 15 - chapter 14

Project Deploy Nexus - Part 2

The final YAML file used in the project can be found here: <https://gitlab.com/twn-devops-bootcamp/latest/15-ansible/ansible-projects/-/blob/main/deploy-nexus.yaml>

Write 3rd play

```
8
9 adduser nexus
10 chown -R nexus:nexus nexus-3.65.0-02
11 chown -R nexus:nexus sonatype-work
12
```

3rd Play

To create nexus user and then we make nexus user the owner of the nexus binary folder and nexus sonatype-work folders.

-> ls -l /opt/

```
root@nexus:~# ls -l /opt/
total 445204
drwxr-xr-x 4 root root      4096 Aug  6 10:03 digitalocean
drwxr-xr-x 6 root root      4096 Aug  6 11:37 nexus
drwxr-xr-x 6 root root      4096 Aug  6 12:22 nexus-3.84.1-01
-rw-r--r-- 1 root root 455869748 Aug  6 11:36 nexus-3.84.1-01-linux-x86_64.tar.gz
drwxr-xr-x 3 root root      4096 Sep 15  2025 sonatype-work
root@nexus:~#
```

Currently the nexus binary folder (nexus and nexus-3.84.1-01 are dupes) and nexus sonatype-work folders are owned by root user.

And we want them to belong to 'nexus' user

```
root@ubuntu-s-1vcpu-1gb-fra1-01:~# ls -l /opt/
total 226444
drwxr-xr-x 4 root root      4096 Feb  6 15:53 digitalocean
drwxr-xr-x 11 root root      4096 Feb  6 16:49 nexus
drwxr-xr-x 10 root root      4096 Feb  6 16:57 nexus-3.65.0-02
-rw-r--r-- 1 root root 231860503 Feb  6 16:15 nexus-3.65.0-02-unix.tar.gz
drwxr-xr-x 3 root root      4096 Feb  6 16:19 sonatype-work
root@ubuntu-s-1vcpu-1gb-fra1-01:~#
```

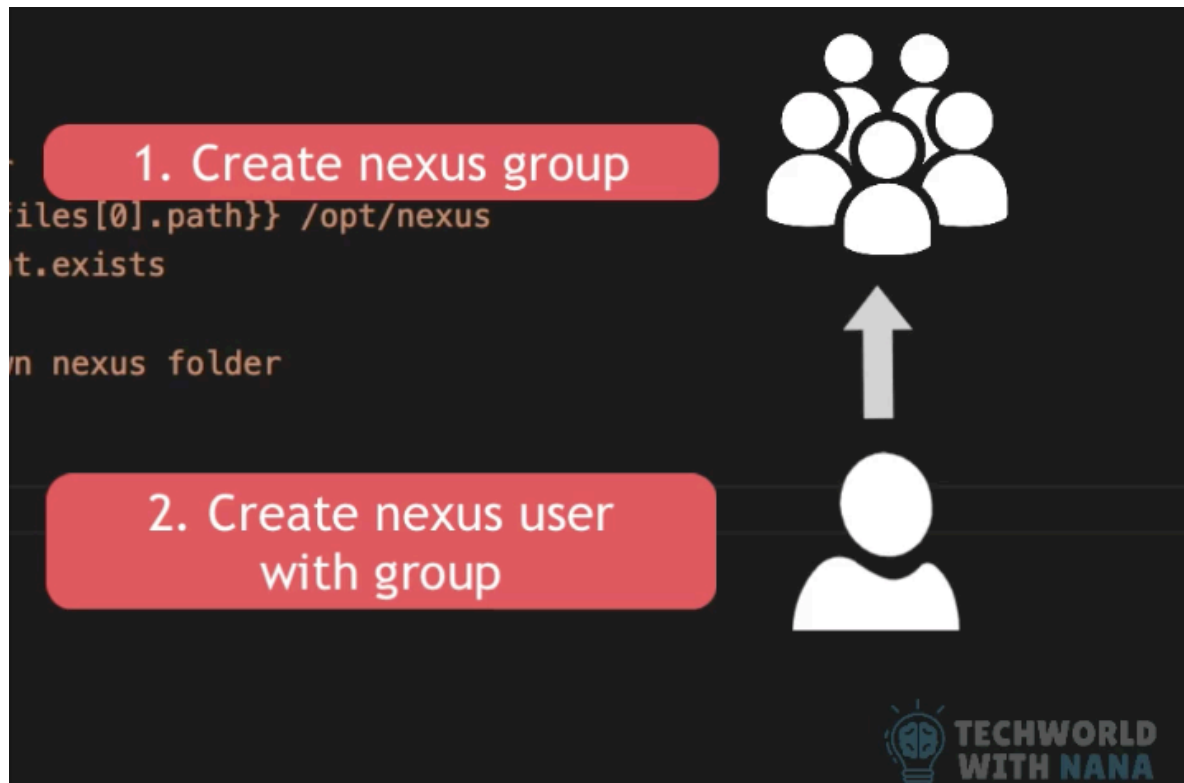


A white silhouette of a person's head and shoulders is shown above a red rounded rectangular button containing the text "nexus user" in white.

First we will write in Ansible the equivalent of -> adduser nexus

But adduser creates the user and also a group, both with one command.

In ansible we need to create the group first, then the user and assign it to the group.

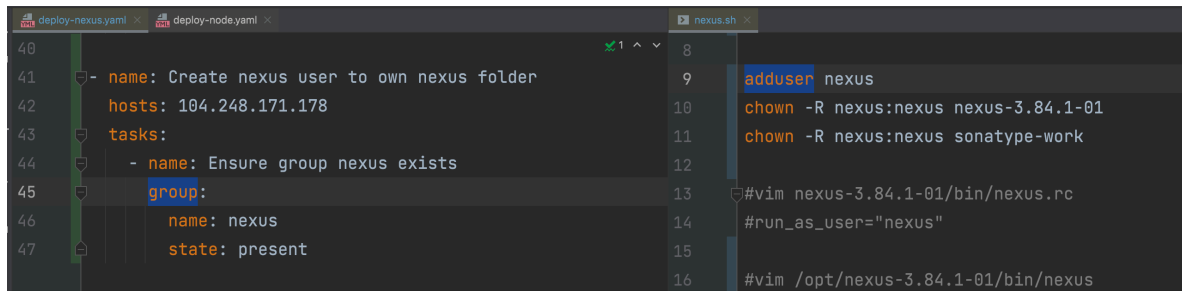


And to create a group we use a module called 'group'

"group" module

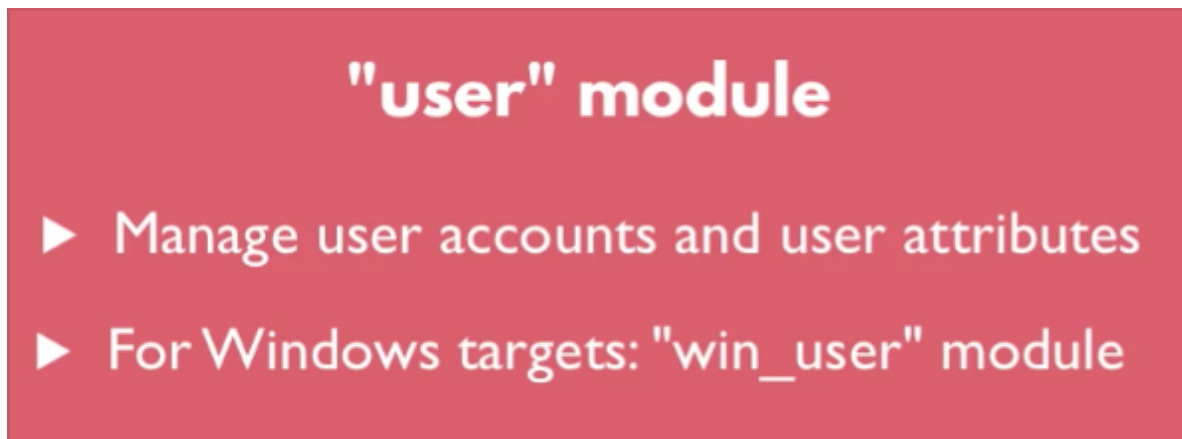
- ▶ Manage presence of groups on a host
- ▶ For Windows targets: "win_group" module

The code so far is:



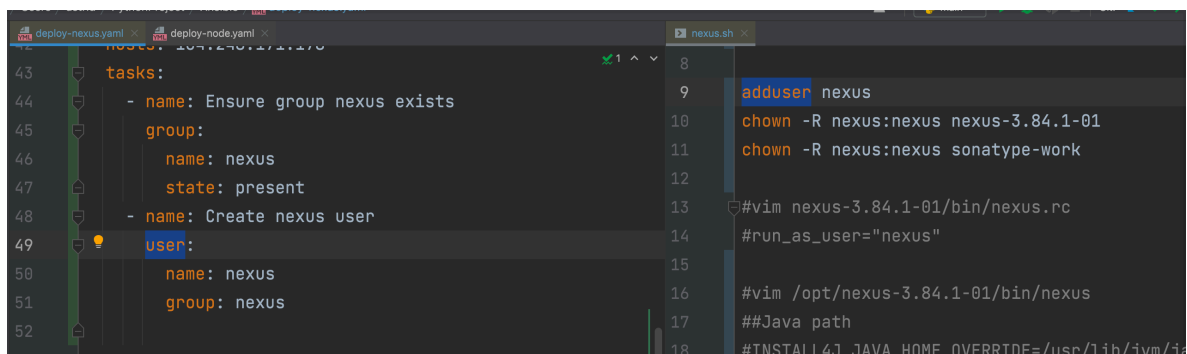
```
deploy-nexus.yml | deploy-node.yml | nexus.sh
40 | 41 | 8
41 | - name: Create nexus user to own nexus folder | 9 | adduser nexus
42 |   hosts: 104.248.171.178 | 10 | chown -R nexus:nexus nexus-3.84.1-01
43 |   tasks: | 11 | chown -R nexus:nexus sonatype-work
44 |     - name: Ensure group nexus exists | 12 |
45 |     group: | 13 | #vim nexus-3.84.1-01/bin/nexus.rc
46 |       name: nexus | 14 | #run_as_user="nexus"
47 |       state: present | 15 |
| | 16 | #vim /opt/nexus-3.84.1-01/bin/nexus
```

Now we create the user using module 'user'



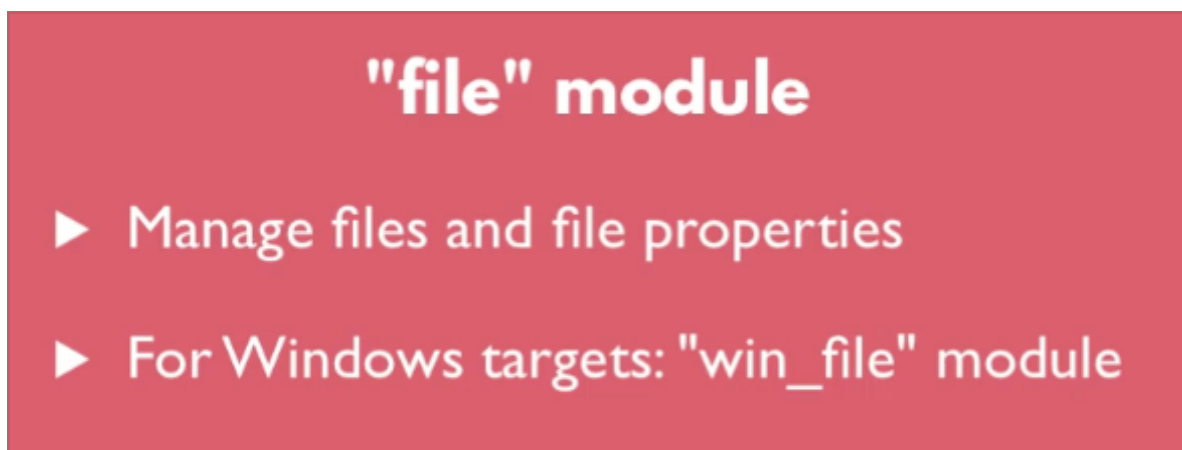
"user" module

- ▶ Manage user accounts and user attributes
- ▶ For Windows targets: "win_user" module



```
deploy-nexus.yml | deploy-node.yml | nexus.sh
43 | 44 | 8
44 |   tasks: | 9 | adduser nexus
45 |     - name: Ensure group nexus exists | 10 | chown -R nexus:nexus nexus-3.84.1-01
46 |     group: | 11 | chown -R nexus:nexus sonatype-work
47 |       name: nexus | 12 |
48 |       state: present | 13 | #vim nexus-3.84.1-01/bin/nexus.rc
49 |     - name: Create nexus user | 14 | #run_as_user="nexus"
50 |     user: | 15 |
51 |       name: nexus | 16 | #vim /opt/nexus-3.84.1-01/bin/nexus
52 |       group: nexus | 17 | ##Java path
| | 18 | #INSTALL4J_JAVA_HOME_OVERRIDE=/usr/lib/jvm/ja
```

And then we can make 'nexus' user the owner of the folders using 'file' module



"file" module

- ▶ Manage files and file properties
- ▶ For Windows targets: "win_file" module

Because we have 2 nexus binary folders, we will use the one that we renamed

'nexus'

```
Users / astrid / PythonProject / Ansible / nexus.sh
deploy-nexus.yaml  deploy-node.yaml  nexus.sh
48 - name: Create nexus user
49   user:
50     name: nexus
51     group: nexus
52 - name: Make nexus user owner of nexus folder
53   file:
54     path: /opt/nexus
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```

Next we specify the state meaning the type which is 'directory'

```
52 - name: Make nexus user owner of nexus folder
53   file:
54     path: /opt/nexus
55     state: directory
56
```

Next we set the owner to 'nexus' and the group to 'nexus' group

```
52 - name: Make nexus user owner of nexus folder
53   file:
54     path: /opt/nexus
55     state: directory
56     owner: nexus
57     group: nexus
58
```

And finally we make it recursive, to change the owner recursively to all the contents of the directory.

-R (recursive) means chown walks into the directory and changes ownership of everything inside it too: subdirectories, files, and their subdirectories, all the way down.

```
52 - name: Make nexus user owner of nexus folder
53   file:
54     path: /opt/nexus
55     state: directory
56     owner: nexus
57     group: nexus
58     recurse: yes
```

Now we don't do the same for sonatype-work

```

59 - name: Make nexus user owner of sonatype-work folder
60   file:
61     path: /opt/sonatype-work
62     state: directory
63     owner: nexus
64     group: nexus
65     recurse: yes
66
10 adduser nexus
11 chown -R nexus:nexus nexus-3.84.1-01
12 chown -R nexus:nexus sonatype-work
13 #vim nexus-3.84.1-01/bin/nexus.rc
14 #run_as_user="nexus"
15
16 #vim /opt/nexus-3.84.1-01/bin/nexus

```

And now let's test the playbook:

-> ansible-playbook -i hosts deploy-nexus.yaml

```

TASK [Make nexus user owner of nexus folder] *****
changed: [104.248.171.178]

TASK [Make nexus user owner of sonatype-work folder] *****
changed: [104.248.171.178]

PLAY RECAP *****
104.248.171.178      : ok=14  changed=2    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

```

It works:

```

root@nexus:~# ls -l /opt/
total 445204
drwxr-xr-x 4 root root      4096 Aug  6 10:03 digitalocean
drwxr-xr-x 6 nexus nexus    4096 Aug  6 11:37 nexus
drwxr-xr-x 6 root root      4096 Aug  6 12:22 nexus-3.84.1-01
-rw-r--r-- 1 root root 455869748 Aug  6 11:36 nexus-3.84.1-01-linux-x86_64.tar.gz
drwxr-xr-x 3 nexus nexus    4096 Sep 15  2025 sonatype-work
root@nexus:~#

```

And now I can switch to the nexus user

-> su - nexus

-> pwd

```

root@nexus:~# su - nexus
$ pwd
/home/nexus
$

```

And I can check the files in the nexus folder and their owners:

-> ls -l /opt/nexus

```
$ ls -l /opt/nexus
total 32
-rw-r--r-- 1 nexus nexus 1022 Sep 15 2025 NOTICE.txt
-rw-r--r-- 1 nexus nexus 11249 Sep 15 2025 OSS-LICENSE.txt
drwxr-xr-x 2 nexus nexus 4096 Aug 6 11:37 bin
drwxr-xr-x 2 nexus nexus 4096 Aug 6 11:37 deploy
drwxr-xr-x 7 nexus nexus 4096 Aug 6 11:37 etc
drwxr-xr-x 3 nexus nexus 4096 Sep 15 2025 jdk
$
```

And the owner of the contents of the nexus binary folder 'nexus' is 'nexus'

Write 4th play

```
12
13 vim nexus-3.65.0-02/bin/nexus.rc
14 run_as_user="nexus"
15
```

4th Play

Nana's nexus folder contains a file called 'nexus.rc' but my folder does not contain it.

This is Nana's

```
root@ubuntu-s-1vcpu-1gb-fra1-01:/# ls /opt/nexus/bin/nexus.rc
```

And this is its content

```
#run_as_user=""
~
~
```

This is mine:

```
root@nexus:~# ls /opt/nexus/bin -a
. . . nexus nexus.vmoptions sonatype-nexus-repository-3.84.1-01.jar
root@nexus:~#
```

I need to create the file.

But for Nanna she needs to edit the contents of the file and for that she used the 'blockinfile' module

"blockinfile" module

- Insert/update/remove a multi-line text surrounded by customizable marker lines

We give it the path of the file that will be edited and then we write the block or multi-line text

In yaml we use the pipe '|' syntax to represent a multi-line string

```
65  - name: Start nexus with nexus user
66    hosts: 134.122.82.225
67    tasks:
68      - name: Set run_as_user nexus
69        blockinfile:
70          path: /opt/nexus/bin/nexus.rc
71          block: |
72            run_as_user="nexus"
73            some_other="sns"
```

But we just need line 72

```
65  - name: Start nexus with nexus user
66    hosts: 134.122.82.225
67    tasks:
68      - name: Set run_as_user nexus
69        blockinfile:
70          path: /opt/nexus/bin/nexus.rc
71          block: |
72            run_as_user="nexus"
```

And Nana runs this playbook and then checks the nexus.rc file:


```
#run_as_user=""  
~  
~
```

And now it uses the 'lineinfile' module

"lineinfile" module

- ▶ Ensures a particular line is in a file, or replace an existing line using regex
- ▶ Useful when you want to change a **single line** in a file only
- ▶ See "replace" module to change multiple lines

And this is the code:

```
65 - name: Start nexus with nexus user  
66   hosts: 134.122.82.225  
67   tasks:  
68     - name: Set run_as_user nexus  
69       lineinfile:  
70         path: /opt/nexus/bin/nexus.rc  
71         line: run_as_user="nexus"
```

And we need to tell the module with line we want to replace and for that we use regex:

```
65 - name: Start nexus with nexus user
66   hosts: 134.122.82.225
67   tasks:
68     - name: Set run_as_user nexus
69       lineinfile:
70         path: /opt/nexus/bin/nexus.rc
71         regexp: '^#run_as_user="" '
72         line: run_as_user="nexus"
```

So we are replacing line 71 with line 72.

And we test it and the file has been edited:


```
67
68 - name: Create nexus.rc and Start nexus with nexus user
69   hosts: 104.248.171.178
70   tasks:
71     copy:
72       dest: /opt/nexus/bin/nexus.rc
73       content: |
74         run_as_user="nexus"
75       owner: nexus
76       group: nexus
77       mode: '0644'
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```

'Copy' module -> copy is already idempotent. Ansible computes a checksum of dest and compares it against what it's about to write; if they match, the task reports ok (unchanged) and does nothing. So as written, re-running the playbook won't rewrite nexus.rc if it already has that exact content.

The nuance: idempotent isn't the same as "never touch it if it exists." If someone hand-edited nexus.rc afterward (different content), copy will overwrite it back to what the task specifies — that's the whole point of copy being declarative/idempotent (it enforces the desired state, not just "create once").

If what you actually want is "only write this file if it doesn't exist yet, and leave it alone forever after that regardless of content" — i.e. respect manual edits — use force: no:

```
- name: Create nexus.rc with run_as_user
  copy:
    dest: /opt/nexus/bin/nexus.rc
    content: |
      run_as_user="nexus"
    owner: nexus
    group: nexus
    mode: '0644'
    force: no
```

```

- name: Create nexus.rc with run_as_user
  copy:
    dest: /opt/nexus/bin/nexus.rc
    content: |
      run_as_user="nexus"
    owner: nexus
    group: nexus
    mode: '0644'
    force: no

```

force: no tells copy to skip the task entirely if dest already exists — no checksum comparison, no overwrite, no when/stat needed.

And 0644 give this permissions:

Who	Digit	Meaning
Owner	6	read + write (4+2)
Group	4	read only
Others	4	read only

I can use 'lineinfile' module instead of 'copy' module.

```

68 - name: Create nexus.rc and Start nexus with nexus user
69 hosts: 104.248.171.178
70 tasks:
71   - name: Ensure run_as_user is set in nexus.rc
72     lineinfile:
73       path: /opt/nexus/bin/nexus.rc
74       regexp: '^#?run_as_user='
75       line: 'run_as_user="nexus"'
76       create: yes
77       owner: nexus
78       group: nexus
79       mode: '0644'

```

```

12
13 vim nexus-3.84.1-01/bin/nexus.rc
14 run_as_user="nexus"
15
16 #vim /opt/nexus-3.84.1-01/bin/nexus
17 ##Java path
18 #INSTALL4J_JAVA_HOME_OVERRIDE=/usr/l3
19
20 #su - nexus
21 #/opt/nexus-3.84.1-01/bin/nexus start
22
23 #ps aux | grep nexus
24 #netstat -lntp

```

- **regexp** searches for a line matching that pattern. The **#?** makes it match whether the line is already commented out (**#run_as_user=...**, which is how some Nexus versions ship this file as a template) or uncommented — either way

it gets replaced with your line: value.

- **line** is what gets written/inserted.
- **create**: yes makes it create the file if it doesn't exist at all (same case as yours, where nexus.rc was missing).
- If regexp finds no match and the file already has other content, lineinfile appends line at the end rather than wiping everything else out.

Trade-off vs copy+content:

- copy replaces the entire file — simplest when you're fully in control of the file and it should only ever contain this one setting.
- lineinfile edits in place — better if the file might pre-exist with other settings/ comments you want preserved, since it won't clobber anything outside the matched line.

For nexus.rc specifically, either works fine since the file's sole purpose is this one setting, but lineinfile is the more defensive choice in case a future Nexus version ships the file with additional commented-out options you'd want left intact.

Refactor the untar task

Now we add 'when' condition to 'unarchive' task because it is being run unnecessarily.

First we moved the 'Check nexus folder stats' task to the top of the tasks (line 17) so we can use it in more places:

```

17 - name: Check nexus folder stats
18   stat:
19     path: /opt/nexus
20   register: stat_result
21 - name: Download Nexus
22   get_url:
23     url: https://download.sonatype.com/nexus/3/nexus-{{nexus_version}}
24     dest: /opt/
25   register: download_result
26 - name: Untar Nexus installer
27   unarchive:
28     src: "{{download_result.dest}}"
29     dest: /opt/
30     remote_src: yes
31   when: not stat_result.stat.exists
32 - name: Find nexus folder
33   find:
34     paths: /opt
35     pattern: "nexus-*"
36     file_type: directory
37   register: find_result
38 - name: Rename nexus folder
39   shell: mv {{find_result.files[0].path}} /opt/nexus
40   when: not stat_result.stat.exists

```

Then we add the 'when' contain to line 31 🙌 , so if 'nexus' folder already exists then don't untar -> skip it.

So I manually remove the nexus binary dupe:

```

root@nexus:~# ls /opt/
digitalocean nexus nexus-3.84.1-01 nexus-3.84.1-01-linux-x86_64.tar.gz sonatype-work
root@nexus:~# rm -rf /opt/nexus-3.84.1-01
root@nexus:~# ls /opt/
digitalocean nexus nexus-3.84.1-01-linux-x86_64.tar.gz sonatype-work
root@nexus:~#

```

And now I only have the nexus binary folder 'nexus'

And re-run the playbook:

-> ansible-playbook -i hosts deploy-nexus.yaml

```

TASK [Download Nexus] *****
ok: [104.248.171.178]

TASK [Untar Nexus installer] *****
skipping: [104.248.171.178]

TASK [Find nexus folder] *****
ok: [104.248.171.178]

TASK [Rename nexus folder] *****
skipping: [104.248.171.178]

PLAY [Create nexus user to own nexus folder] *****

TASK [Gathering Facts] *****
ok: [104.248.171.178]

```

It skipped the untar task this time, so it skipped two tasks (the other task is renaming the folder)

```

TASK [Gathering Facts] *****
ok: [104.248.171.178]

TASK [Ensure run_as_user is set in nexus.rc] *****
changed: [104.248.171.178]

PLAY RECAP *****
104.248.171.178 : ok=15  changed=2  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0

```

And it says that there was a change regarding nexus.rc, so let's check the droplet

-> ls /opt/nexus/bin -l

```

root@nexus:~# ls /opt/nexus/bin -l
total 281604
-rwxr-xr-x 1 nexus nexus      6735 Sep 15  2025 nexus
-rw-r--r-- 1 nexus nexus        20 Aug  6  18:32 nexus.rc
-rw-r--r-- 1 nexus nexus     1325 Sep 15  2025 nexus.vmoptions
-rw-r--r-- 1 nexus nexus 288340658 Sep 15  2025 sonatype-nexus-repository-3.84.1-01.jar
root@nexus:~#

```

Nexus.rc now is there and now let's look inside

-> cat /opt/nexus/bin/nexus.rc

```

root@nexus:~# cat /opt/nexus/bin/nexus.rc
run_as_user="nexus"
root@nexus:~#

```

Last step for this play we need to run nexus:


```

68 - name: Start nexus with nexus user
69   hosts: 104.248.171.178
70   tasks:
71     - name: Ensure run_as_user is set in nexus.rc
72       lineinfile:
73         path: /opt/nexus/bin/nexus.rc
74         regexp: '^#?run_as_user='
75         line: 'run_as_user="nexus"'
76         create: yes
77         owner: nexus
78         group: nexus
79         mode: '0644'
80     - name: Start nexus
81       command: /opt/nexus/bin/nexus start
82
11 #chown -R nexus:nexus sonatype-work
12
13 vim nexus-3.84.1-01/bin/nexus.rc
14 run_as_user="nexus"
15 su - nexus
16 /opt/nexus-3.84.1-01/bin/nexus start
17
18 #vim /opt/nexus-3.84.1-01/bin/nexus
19 ##Java path
20 #INSTALL4J_JAVA_HOME_OVERRIDE=/usr/lib/jvm/j
21
22
23
24 #ps aux | grep nexus
25 #netstat -lntp

```

But we are missing switching the user to 'nexus'

```

67
68 - name: Start nexus with nexus user
69   hosts: 104.248.171.178
70   become: True
71   become_user: nexus
72   tasks:
73     - name: Ensure run_as_user is set in nexus.rc
74       lineinfile:
75         path: /opt/nexus/bin/nexus.rc
76         regexp: '^#?run_as_user='
77         line: 'run_as_user="nexus"'
78         create: yes
79         owner: nexus
80         group: nexus
81         mode: '0644'
82     - name: Start nexus
83       command: /opt/nexus/bin/nexus start
84
11 #chown -R nexus:nexus sonatype-work
12
13 vim nexus-3.84.1-01/bin/nexus.rc
14 run_as_user="nexus"
15 su - nexus
16 /opt/nexus-3.84.1-01/bin/nexus start
17
18 #vim /opt/nexus-3.84.1-01/bin/nexus
19 ##Java path
20 #INSTALL4J_JAVA_HOME_OVERRIDE=/usr/lib/jvm/java-1
21
22
23
24 #ps aux | grep nexus
25 #netstat -lntp

```

And now I don't need lines 79 and 80 as owner and group are redundant

```

68 - name: Start nexus with nexus user
69   hosts: 104.248.171.178
70   become: True
71   become_user: nexus
72   tasks:
73     - name: Ensure run_as_user is set in nexus.rc
74       lineinfile:
75         path: /opt/nexus/bin/nexus.rc
76         regexp: '^#?run_as_user='
77         line: 'run_as_user="nexus"'
78         create: yes
79         # owner: nexus
80         # group: nexus
81         mode: '0644'
82     - name: Start nexus
83       command: /opt/nexus/bin/nexus start

```

I remove the nexus.rc from my droplet

```

root@nexus:~# rm /opt/nexus/bin/nexus.rc
root@nexus:~# ls /opt/nexus/bin -l
total 281600
-rwxr-xr-x 1 nexus nexus      6735 Sep 15  2025 nexus
-rw-r--r-- 1 nexus nexus      1325 Sep 15  2025 nexus.vmoptions
-rw-r--r-- 1 nexus nexus 288340658 Sep 15  2025 sonatype-nexus-repository-3.84.1-01.jar

```

And let's execute the playbook

-> ansible-playbook -i hosts deploy-nexus.yaml

```

TASK [Ensure run_as_user is set in nexus.rc] *****
changed: [104.248.171.178]

TASK [Start nexus] *****
changed: [104.248.171.178]

PLAY RECAP *****
104.248.171.178      : ok=16  changed=2    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

```

And check the droplet

```

root@nexus:~# ls /opt/nexus/bin -l
total 281604
-rwxr-xr-x 1 nexus nexus      6735 Sep 15  2025 nexus
-rw-r--r-- 1 nexus nexus       20 Aug  6 18:48 nexus.rc
-rw-r--r-- 1 nexus nexus     1325 Sep 15  2025 nexus.vmoptions
-rw-r--r-- 1 nexus nexus 288340658 Sep 15  2025 sonatype-nexus-repository-3.84.1-01.jar
root@nexus:~#

```

Nexus.rc is created and with the right owner and group 🤔

Now let's check if nexus is running

-> ps aux | grep nexus

```

root@nexus:~# ps aux | grep nexus
root      25115  0.0  0.2  7076  2164 pts/0    S+   18:52   0:00 grep --color=auto nexus
root@nexus:~#

```

Nexus is not running in the droplet ❌

So even ansible says it worked, that the task to start nexus didn't failed. In reality it failed.

So we always should validate to check that the app started successfully, like we did in 'deploy-node.yaml'

```

43
44 - name: Ensure app is running
45   shell: ps aux | grep node
46   register: app_status
47 - debug: msg={{app_status.stdout_lines}}
48

```

The other issue is that the droplet doesn't have enough space, it is too small and we can see the actual logs if we run nexus manually in the droplet

-> /opt/nexus/bin/nexus run

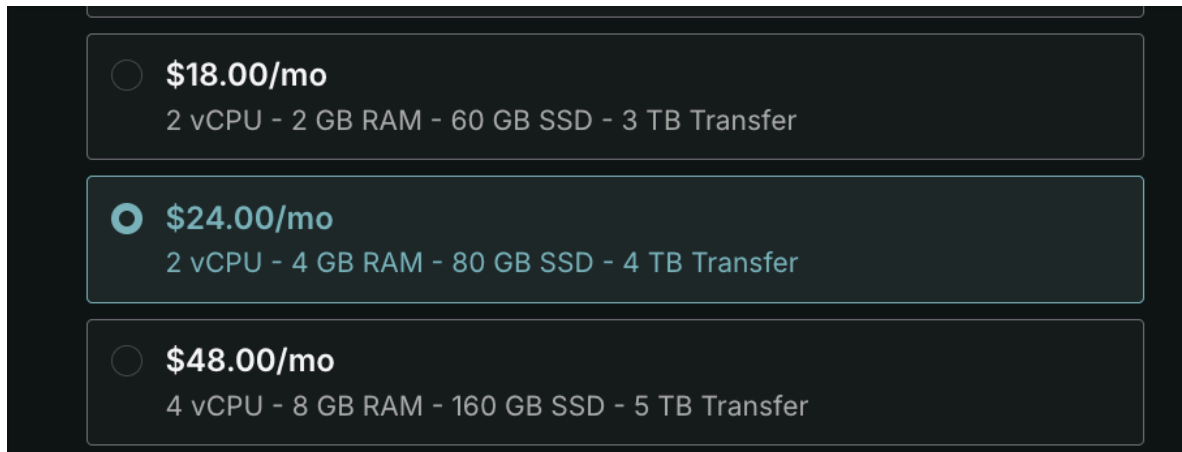
```

root@nexus:~# /opt/nexus/bin/nexus run
OpenJDK 64-Bit Server VM warning: INFO: os::commit_memory(0x000000078f550000, 1890254848, 0) failed; error='Not enough space' (errno=12)
#
# There is insufficient memory for the Java Runtime Environment to continue.
# Native memory allocation (mmap) failed to map 1890254848 bytes. Error detail: committing reserved memory.
# An error report file with more information is saved as:
# /opt/nexus/hs_err_pid25144.log
root@nexus:~#

```

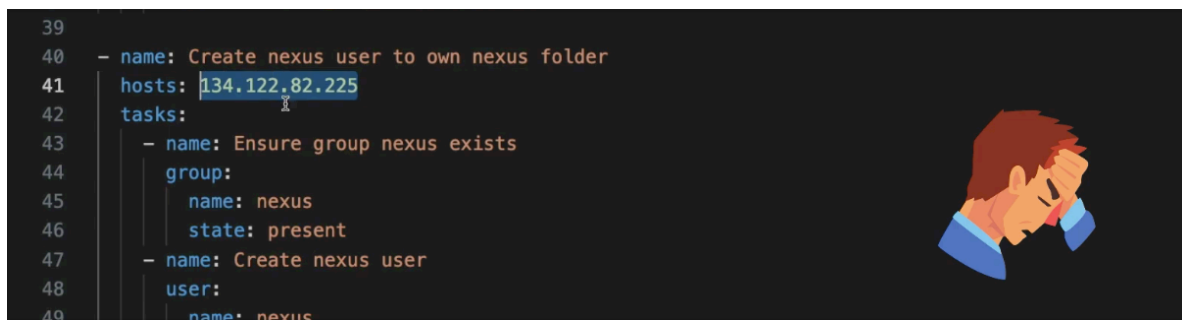
Insufficient memory 🙅

So we need to destroy this droplet and create a new bigger droplet and install nexus there 💡

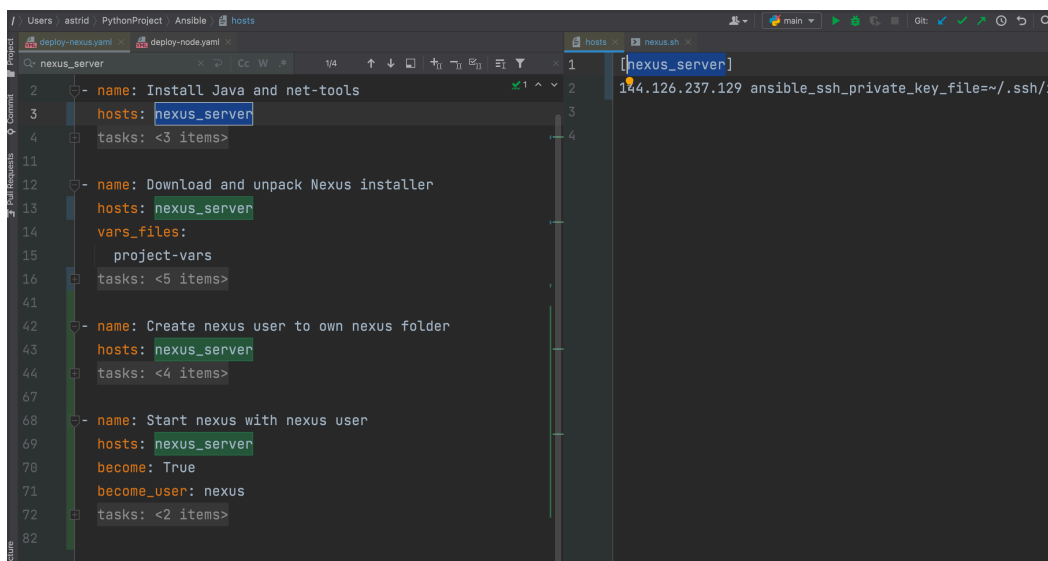


Because we have our ansible with all script it out destroying and creating a new droplet and then getting the new droplet configured is easy ❤️

Only thing is that our ansible playbook needs the IP address of the previous droplet and now it needs to be updated in many places



The solution is to always name the servers even if there is only one!



Run the playbook

-> ansible-playbook -i hosts deploy-nexus.yml

```
TASK [Ensure run_as_user is set in nexus.rc] *****
changed: [167.172.63.164]

TASK [Start nexus] *****
changed: [167.172.63.164]

PLAY RECAP *****
167.172.63.164      : ok=18  changed=12  unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

And we check the new droplet

```
root@nexus:~# ps aux | grep nexus
nexus  9235  176  20.5 5449228 824176 ?        Ssl  19:18   0:23 /opt/nexus/jdk/temurin_17.0.13_11_linux_x86_64/jdk-17.0.13+11/bin/java -se
rver -Dnexus.installer.type=linux-x86-64 -Xms2703m -Xmx2703m -XX:+UnlockDiagnosticVMOptions -XX:+LogVMOutput -XX:LogFile=../sonatype-work/ne
us3/log/jvm.log -XX:-OmitStackTraceInFastThrow -Dkaraf.home=. -Dkaraf.base=. -Djava.util.logging.config.file=etc/spring/java.util.logging.pro
perties -Dkaraf.data=../sonatype-work/nexus3 -Dkaraf.log=../sonatype-work/nexus3/log -Djava.io.tmpdir=../sonatype-work/nexus3/tmp -Djdk.tls.e
phemeralDHKeySize=2048 -Dfile.encoding=UTF-8 --add-reads=java.xml=java.logging --add-opens java.base/java.security=ALL-UNNAMED --add-opens ja
va.base/java.net=ALL-UNNAMED --add-opens java.base/java.lang=ALL-UNNAMED --add-opens java.base/java.util=ALL-UNNAMED --add-opens java.naming/
javax.naming.spi=ALL-UNNAMED --add-opens java.rmi/sun.rmi.transport.tcp=ALL-UNNAMED --add-exports=java.base/sun.net.www.protocol.http=ALL-UNN
AMED --add-exports=java.base/sun.net.www.protocol.https=ALL-UNNAMED --add-exports=java.base/sun.net.www.protocol.jar=ALL-UNNAMED --add-export
s=jdk.xml.dom/org.w3c.dom.html=ALL-UNNAMED --add-exports=jdk.naming.rmi/com.sun.jndi.url.rmi=ALL-UNNAMED --add-exports=java.security.sasl/com
.sun.security.sasl=ALL-UNNAMED --add-exports=java.base/sun.security.x509=ALL-UNNAMED --add-exports=java.base/sun.security.rsa=ALL-UNNAMED --a
dd-exports=java.base/sun.security.pkcs=ALL-UNNAMED -jar /opt/nexus/bin/sonatype-nexus-repository-3.84.1-01.jar
root    9263   0.0  0.0   7076   2156 pts/0    S+   19:18   0:00 grep --color=auto nexus
root@nexus:~#
```

And nexus is running 🤖

Write 5th play

```
18
19  ps aux | grep nexus
20  netstat -lnt
```

5th Play

So we log the stats of the nexus app so we don't have to ssh into the droplet to see if it is working.

```
83  name: Verify nexus running
84  hosts: nexus_server
85  tasks:
86  - name: Check with ps
87    shell: ps aux | grep nexus
88    register: app_status
```

```
15  #su - nexus
16  #/opt/nexus-3.84.1-01/bin/t
17
18  ps aux | grep nexus
19  netstat -lnt
```

And we add the task to get the stats

```
83  - name: Verify nexus running
84    hosts: nexus_server
85    tasks:
86      - name: Check with ps
87        shell: ps aux | grep nexus
88        register: app_status
89        debug: msg={{app_status.stdout_lines}}
90      - name: Check with netstat
91        shell: netstat -plnt
92        register: app_status
93        debug: msg={{app_status.stdout_lines}}
11  #chown -R nexus:nexus sonatype-work
12
13  #vim nexus-3.84.1-01/bin/nexus.rc
14  #run_as_user="nexus"
15  #su - nexus
16  #/opt/nexus-3.84.1-01/bin/nexus start
17
18  ps aux | grep nexus
19  netstat -lnpt
```

Q:

netstat -plnt or netstat -lnpt?

A:

Both are identical — order of combined single-letter flags doesn't matter to netstat (or almost any Unix CLI using standard option parsing). -plnt and -lnpt just bundle the same four flags in a different sequence:

- -p — show the PID/program name owning each socket
- -l — only listening sockets
- -n — numeric output (skip DNS/service-name resolution, faster and avoids hangs)
- -t — TCP only

So netstat -plnt and netstat -lnpt produce exactly the same output. Use whichever reads more naturally to you.

I kill the nexus process

-> kill 9235

```
root@nexus:~# kill 9235
root@nexus:~# ps aux | grep nexus
root      9398  0.0  0.0  7076  2164 pts/0    S+   19:29   0:00 grep --color=auto nexus
root@nexus:~#
```

And execute the playbook

-> ansible-playbook -i hosts deploy-nexus.yaml

```
TASK [debug] *****
ok: [167.172.63.164] => {
  "msg": [
    "nexus      10281  154 56.0 5637992 2245556 ?      SL   19:34   1:58 /opt/ne
17.0.13+11/bin/java -server -Dnexus.installer.type=linux-x86-64 -Xms2703m -Xmx2703m
tput -XX:LogFile=../sonatype-work/nexus3/log/jvm.log -XX:-OmitStackTraceInFastThrow
ogging.config.file=etc/spring/java.util.logging.properties -Dkaraf.data=../sonatype
xus3/log -Djava.io.tmpdir=../sonatype-work/nexus3/tmp -Djdk.tls.ephemeralDHKeySize=
  ]
}
```

We get the first log, show nexus is running.

```
TASK [Check with netstat] *****
changed: [167.172.63.164]

TASK [debug] *****
ok: [167.172.63.164] => {
  "msg": [
    "Active Internet connections (only servers)",
    "Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name    ",
    "tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      1/init               ",
    "tcp        0      0 0.0.0.0:53              0.0.0.0:*               LISTEN      711/systemd-resolve ",
    "tcp6       0      0 :::8081                 :::*                    LISTEN      10281/java           ",
    "tcp6       0      0 :::22                   :::*                    LISTEN      1/init               "
  ]
}
```

And we get the netstat logs!

And I check if nexus is listening but how do I know if nexus is listening?

A:

You infer it from two clues, not a literal name — netstat only tells you the process name as the OS sees it (java), not what that JVM is actually running.

- Port 8081 — that's Nexus Repository Manager's default HTTP port.

Seeing :::8081 ... LISTEN 11297/java is the strong hint.

- Process is java — Nexus is a Java application, so it shows up as java, not nexus, in the process list. That's expected and doesn't mean anything's wrong.

To actually confirm PID 11297 is Nexus (rather than some other Java app that happens to be on 8081), check what that process is running:

```
ps -p 11297 -f
```

```
# or
```

```
cat /proc/11297/cmdline | tr '\0' ' '
```

You should see the classpath/jar referencing sonatype-nexus-repository or the nexus.vmoptions config path in the command line.

```

root@nexus:~# cat /proc/11297/cmdline | tr '\0' ' '
/opt/nexus/jdk/temurin_17.0.13_11_linux_x86_64/jdk-17.0.13+11/bin/java -server -Dnexus.installer.type=linux-x86-64 -Xms2703m -Xmx2703m -XX:+U
nlockDiagnosticVMOptions -XX:+LogVMOutput -XX:LogFile=./sonatype-work/nexus3/log/jvm.log -XX:-OmitStackTraceInFastThrow -Dkaraf.home=. -Dkar
af.base=. -Djava.util.logging.config.file=etc/spring/java.util.logging.properties -Dkaraf.data=./sonatype-work/nexus3 -Dkaraf.log=./sonatyp
e-work/nexus3/log -Djava.io.tmpdir=./sonatype-work/nexus3/tmp -Djdk.tls.ephemeralDHKeySize=2048 -Dfile.encoding=UTF-8 --add-reads=java.xml=
java.logging --add-opens java.base/java.security=ALL-UNNAMED --add-opens java.base/java.net=ALL-UNNAMED --add-opens java.base/java.lang=ALL-UN
NAMED --add-opens java.base/java.util=ALL-UNNAMED --add-opens java.naming/javax.naming.spi=ALL-UNNAMED --add-opens java.rmi/sun.rmi.transport
.tcp=ALL-UNNAMED --add-exports=java.base/sun.net.www.protocol.http=ALL-UNNAMED --add-exports=java.base/sun.net.www.protocol.https=ALL-UNNAMED
--add-exports=java.base/sun.net.www.protocol.jar=ALL-UNNAMED --add-exports=jdk.xml.dom/org.w3c.dom.html=ALL-UNNAMED --add-exports=jdk.naming
.rmi/com.sun.jndi.url.rmi=ALL-UNNAMED --add-exports=java.security.sasl/com.sun.security.sasl=ALL-UNNAMED --add-exports=java.base/sun.security
.x509=ALL-UNNAMED --add-exports=java.base/sun.security.rsa=ALL-UNNAMED --add-exports=java.base/sun.security.pkcs=ALL-UNNAMED -jar /opt/nexus/
bin/sonatype-nexus-repository-3.84.1-01.jar root@nexus:~#

```

Or just hit the actual service:

```
curl -sI http://localhost:8081
```

A response (even a redirect to the login page) confirms Nexus is up and serving on that port — that's the most direct proof, since it tests the actual application, not just "something is listening."

```

root@nexus:~# curl -sI http://localhost:8081
HTTP/1.1 200 OK
Server: Nexus/3.84.1-01 (COMMUNITY)
X-Content-Type-Options: nosniff
Content-Security-Policy: default-src http: data: blob: 'unsafe-inline'; script-src http: 'unsafe-inline' 'unsafe-eval'
X-Frame-Options: DENY
X-XSS-Protection: 1; mode=block
Content-Type: text/html
Last-Modified: Thu, 06 Aug 2026 19:49:48 GMT
Pragma: no-cache
Cache-Control: no-cache, no-store, max-age=0, must-revalidate, post-check=0, pre-check=0
Expires: 0
Content-Length: 8263

root@nexus:~#

```

So in my case, nexus is listening 

Nana's log is incomplete we can't see the nexus port listening yet which is 8081

```

TASK [debug] *****
ok: [64.226.94.75] => {
  "msg": [
    "Active Internet connections (only servers)",
    "Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name  ",
    "tcp        0      0 127.0.0.1:37193         0.0.0.0:*               LISTEN      5848/java          ",
    "tcp        0      0 127.0.0.0.54:53        0.0.0.0:*               LISTEN      604/systemd-resolve ",
    "tcp        0      0 127.0.0.0.53:53        0.0.0.0:*               LISTEN      604/systemd-resolve ",
    "tcp6       0      0 :::22                  :::*                    LISTEN      1/init             "
  ]
}

```

Nana checked the droplet and check the netstat
-> netstat -plnt


```

root@ubuntu-s-4vcpu-8gb-fra1-01:~# netstat -plnt
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:37193         0.0.0.0:*               LISTEN      5848/java
tcp        0      0 127.0.0.54:53          0.0.0.0:*               LISTEN      604/systemd-resolve
tcp        0      0 0.0.0.0:8081           0.0.0.0:*               LISTEN      5848/java
tcp        0      0 127.0.0.53:53          0.0.0.0:*               LISTEN      604/systemd-resolve
tcp6       0      0 :::22                  :::*                   LISTEN      1/init
root@ubuntu-s-4vcpu-8gb-fra1-01:~#

```

And there she can see 8081 port to access nexus from a browser.

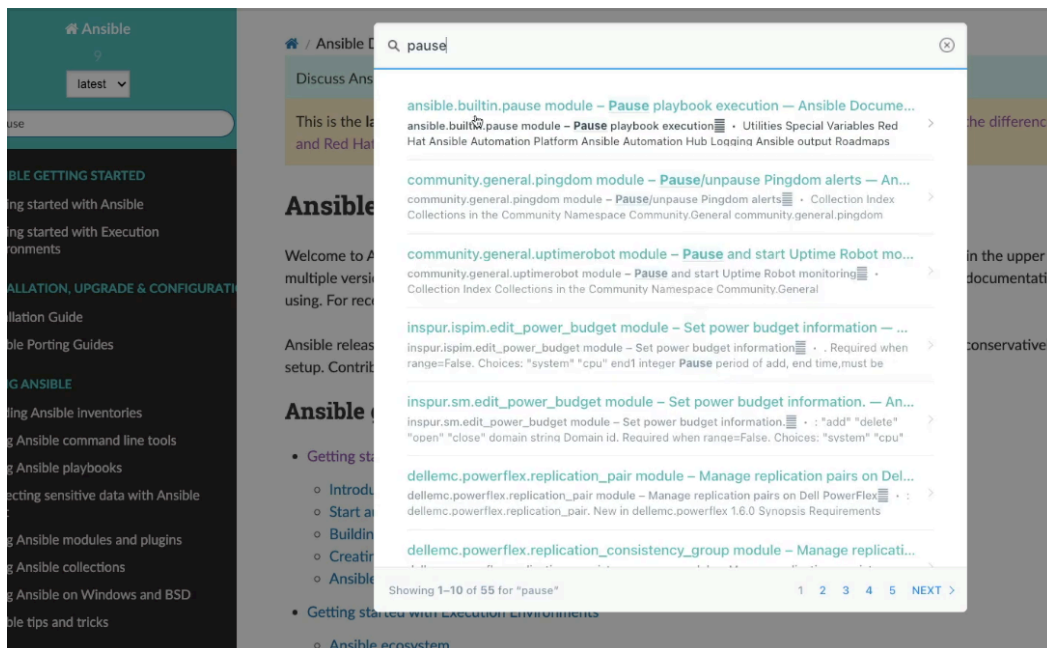
The reason for that difference is that it takes time for the app to be ready and we are getting the stats right away after we started the app, so we need to wait bit before calling the status.

-> timing issue

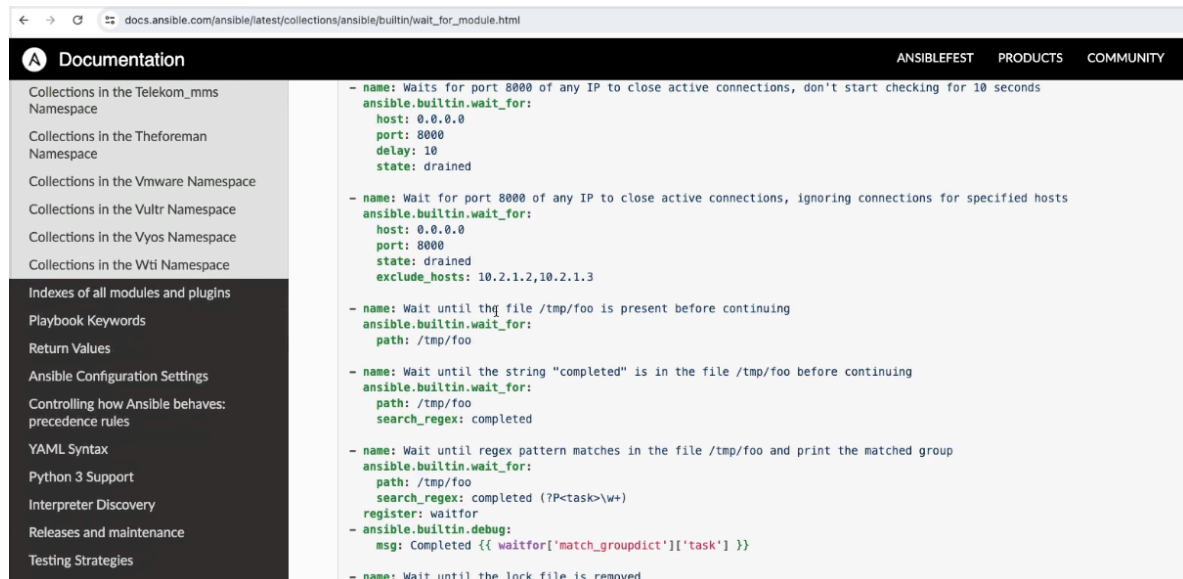
So we can configure our play to wait before running the netstat task.

We can use two modules:

- 'Pause'



- 'wait_for'

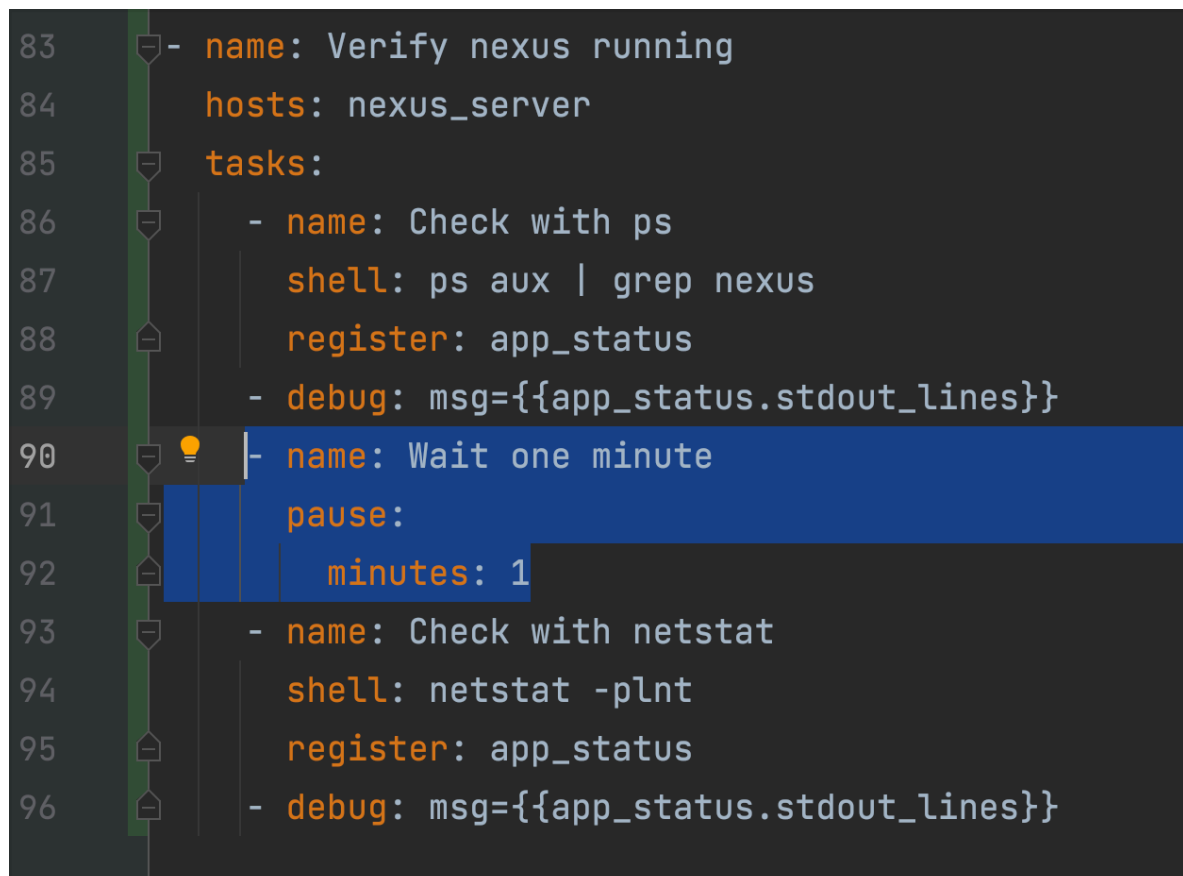


The screenshot shows the Ansible documentation for the `wait_for` module. The left sidebar contains a navigation menu with links to various namespaces and modules. The main content area displays several examples of the `wait_for` module usage:

- `- name: Waits for port 8000 of any IP to close active connections, don't start checking for 10 seconds`
`ansible.builtin.wait_for:`
`host: 0.0.0.0`
`port: 8000`
`delay: 10`
`state: drained`
- `- name: Wait for port 8000 of any IP to close active connections, ignoring connections for specified hosts`
`ansible.builtin.wait_for:`
`host: 0.0.0.0`
`port: 8000`
`state: drained`
`exclude_hosts: 10.2.1.2,10.2.1.3`
- `- name: Wait until the file /tmp/foo is present before continuing`
`ansible.builtin.wait_for:`
`path: /tmp/foo`
- `- name: Wait until the string "completed" is in the file /tmp/foo before continuing`
`ansible.builtin.wait_for:`
`path: /tmp/foo`
`search_regex: completed`
- `- name: Wait until regex pattern matches in the file /tmp/foo and print the matched group`
`ansible.builtin.wait_for:`
`path: /tmp/foo`
`search_regex: completed (?P<task>\w+)`
`register: waitfor`
`- ansible.builtin.debug:`
`msg: Completed {{ waitfor['match_groupdict']['task'] }}`
- `- name: Wait until the lock file is removed`

Where we can wait for something to happen and then we continue, like wait until we get a port.

We will use 'pause' module in this chapter and we will wait for 1 minute



```
83 - name: Verify nexus running
84   hosts: nexus_server
85   tasks:
86     - name: Check with ps
87       shell: ps aux | grep nexus
88       register: app_status
89     - debug: msg={{app_status.stdout_lines}}
90     - name: Wait one minute
91       pause:
92         minutes: 1
93     - name: Check with netstat
94       shell: netstat -plnt
95       register: app_status
96     - debug: msg={{app_status.stdout_lines}}
```

If the port hasn't open within one minute it means the app has some problems



**Everything is
automated!**

